

AIM:

Upload and analyse the given dataset and perform data preprocessing

PYTHON CODE:

```
import pandas as pd
import numpy as np

df = pd.read_excel("preprocessed_dataset.xlsx")

print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Records:")
print(df.duplicated().sum())

df.drop_duplicates(inplace=True)

numerical_columns = df.select_dtypes(include=np.number).columns
categorical_columns = df.select_dtypes(include="object").columns

for column in numerical_columns:
    df[column] = df[column].fillna(df[column].mean())

for column in categorical_columns:
    if df[column].isnull().any():
        df[column] = df[column].fillna(df[column].mode()[0])

print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())

print("\nCleaned Dataset:")
print(df.head())

df.to_excel("preprocessed_dataset.xlsx", index=False)
```

OUTPUT:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
		Age_Group.1						
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000

Dataset Shape:

(10, 9)

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
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0	CustomerID	10 non-null	int64
1	Age_Group	10 non-null	str
2	Rating(1-5)	10 non-null	int64
3	Hotel	10 non-null	str
4	FoodPreference	10 non-null	str
5	Bill	10 non-null	int64
6	NoOfPax	10 non-null	int64
7	EstimatedSalary	10 non-null	int64
8	Age_Group.1	10 non-null	str

dtypes: int64(5), str(4)

memory usage: 852.0 bytes

Missing Values:

CustomerID 0

Age_Group 0

Rating(1-5) 0

Hotel 0

FoodPreference 0

Bill 0

NoOfPax 0

EstimatedSalary 0

Age_Group.1 0

dtype: int64

Duplicate Records:

0

/Users/yuva/Desktop/Code_Hub/FDS_Exp/Exp_3/exp4.py:23: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This

behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.

See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
categorical_columns = df.select_dtypes(include="object").columns
```

Missing Values After Preprocessing:

```
CustomerID      0
Age_Group       0
Rating(1-5)     0
Hotel           0
FoodPreference  0
Bill            0
NoOfPax         0
EstimatedSalary 0
Age_Group.1     0
dtype: int64
```

Cleaned Dataset:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000

RESULT:

The given dataset is analysed and preprocessed successfully.